

Quantitative Convergence Rates for the Kuramoto Self-Consistency Iteration

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Abstract

We establish explicit convergence rates for the self-consistency iteration of the Kuramoto model on the Ott–Antonsen (OA) manifold. At critical coupling $K = K_c$, we prove that iterates of the self-consistency map Φ satisfy $r_n \leq r_0/\sqrt{1+Bn}$ where $B = K^3 r_0^2 / (\gamma_{\max}(2\gamma_{\max} + K)^2)$, providing the first explicit bound with computable constants for the critical regime. Above K_c , for initial data r_0 with $\Phi'(r_0) < 1$, we prove geometric convergence with rate $L_0 = \Phi'(r_0)$. For the real scalar reduction we construct an L^2 Lyapunov function and prove stability via a new sum-of-squares pair bound inequality. For the general complex Ott–Antonsen equation, we establish the Dietert energy identity $d\Psi/dt = K|\eta|^2$ and obtain stability via two explicit axioms encoding the Dietert–Fernandez Landau damping theorem. All results are formally verified in Lean 4 with the Mathlib library (87 files, 0 `sorry`, 2 axioms).

1 Introduction

The Kuramoto model [6, 7] describes the dynamics of N coupled phase oscillators:

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N, \quad (1)$$

where ω_i are natural frequencies drawn from a distribution g and $K > 0$ is the coupling strength. The central phenomenon is a phase transition at a critical coupling K_c : for $K < K_c$ the oscillators are incoherent, while for $K > K_c$ a macroscopic fraction synchronizes, producing a nonzero order parameter $r^* > 0$.

Since Kuramoto’s original work, the mathematical theory has advanced through several stages: the self-consistency equation [7, 11], the Ott–Antonsen reduction [8, 9], bifurcation analysis [2], and Landau damping for the kinetic PDE [4, 5]. While existence and uniqueness of r^* are well-established (see [1]), explicit convergence rates for the Picard iteration $r_{n+1} = \Phi(r_n)$ —including computable constants—have not appeared in the literature. The critical dynamics at $K = K_c$, where $\Phi'(r^*) \rightarrow 1$, have been studied only phenomenologically [10].

In this paper we establish three main results:

- (i) **Critical algebraic decay** (Theorem 3.2): At $K = K_c$, the iterates $r_n = \Phi^n(r_0)$ satisfy $r_n \leq r_0/\sqrt{1+Bn}$ with an explicit constant B depending on K , r_0 , and $\gamma_{\max}$, giving $O(1/\sqrt{n})$ decay with a computable prefactor.
- (ii) **Geometric convergence** (Theorem 4.1): For $K > K_c$ and $r_0 \in (0, r^*)$ with $\Phi'(r_0) < 1$, the iterates converge geometrically: $r^* - r_n \leq L_0^n(r^* - r_0)$ where $L_0 = \Phi'(r_0)$.
- (iii) **OA stability** (Theorem 5.4): On the real scalar OA manifold, $V = \int (z_\omega - z_\omega^*)^2 g d\mu$ is a Lyapunov function via a new sum-of-squares pair bound inequality. For the general complex OA, stability follows from a single explicit axioms encoding the Dietert–Fernandez theorem.

All proofs are formally verified in the Lean 4 proof assistant. The formalization is available at <https://github.com/taejun-song/kuramoto-lean>.

1.1 Relation to prior work

The $O(1/\sqrt{n})$ rate at criticality is the natural consequence of the cubic tangency $\Phi(r) = r - cr^3 + O(r^5)$ at $K = K_c$, a structure implicit in Crawford’s center manifold analysis [3]. Our contribution is making this quantitative: we provide explicit, distribution-dependent constants and a complete inductive proof that has been machine-checked. The reciprocal-growth technique we employ is standard for maps with cubic tangency (see e.g. [11]), but has not been applied to the Kuramoto self-consistency map with explicit bounds.

The OA reduction [8] was developed for Lorentzian distributions and later extended to rational frequency densities. Our L^2 Lyapunov analysis on the real scalar OA works for arbitrary symmetric distributions on a probability space (Ω, μ) , not requiring analyticity of g . The sum-of-squares pair bound inequality (Lemma 5.2) establishing $dV/dt \leq 0$ appears to be new as a proof technique for OA stability.

The connection to the full Kuramoto PDE relies on the Landau damping results of Dietert and Fernandez [4], which we incorporate as two precisely stated axioms.

2 The Self-Consistency Map

2.1 Setup and equilibrium formula

Let (Ω, μ) be a probability space and $\gamma : \Omega \rightarrow \mathbb{R}_{>0}$ a measurable function representing the absolute values of natural frequencies (equivalently, the half-widths in the symmetric case $g(\omega) = g(-\omega)$, where only $|\omega|$ enters the self-consistency equation). We assume throughout that γ is essentially bounded:

$$\gamma_{\min} := \text{ess inf}_\omega \gamma(\omega) > 0, \quad \gamma_{\max} := \text{ess sup}_\omega \gamma(\omega) < \infty.$$

The *critical coupling* is

$$K_c = \frac{2}{\int_\Omega \frac{1}{\gamma(\omega)} d\mu(\omega)}. \quad (2)$$

For $K > 0$ and $r > 0$, the individual equilibrium amplitude satisfies $(K/2)r(1-\alpha^2) = \gamma\alpha$, i.e., $(K/2)r \cdot \alpha^2 + \gamma \cdot \alpha - (K/2)r = 0$. We use the *rationalized form* of the positive root:

Definition 2.1 (Equilibrium amplitude).

$$\alpha^*(\gamma, K, r) = \frac{Kr}{\gamma + \sqrt{\gamma^2 + K^2 r^2}}.$$

This presentation, equivalent to the classical formula $\alpha^* = (-\gamma + \sqrt{\gamma^2 + K^2 r^2})/(Kr)$ after rationalization, is the algebraic engine enabling all subsequent estimates.

Proposition 2.2 (Properties of α^*). *For $\gamma, K, r > 0$:*

- (a) $0 < \alpha^*(\gamma, K, r) < 1$;
- (b) α^* is strictly decreasing in γ ;
- (c) $\frac{Kr}{2\sqrt{\gamma^2 + K^2 r^2}} \leq \alpha^*(\gamma, K, r) < \frac{Kr}{\gamma}$;
- (d) α^* is strictly increasing in r .

The self-consistency map is

$$\Phi(r) = \int_{\Omega} \alpha^*(\gamma(\omega), K, r) d\mu(\omega). \quad (3)$$

A fixed point $\Phi(r^*) = r^*$ with $r^* > 0$ exists if and only if $K > K_c$.

2.2 Slope estimates

Theorem 2.3 (Slope of Φ). *For any $r > 0$, the derivative of Φ is*

$$\Phi'(r) = \int_{\Omega} \frac{K\gamma(\omega)}{\sqrt{\gamma(\omega)^2 + K^2 r^2}(\gamma(\omega) + \sqrt{\gamma(\omega)^2 + K^2 r^2})} d\mu(\omega). \quad (4)$$

Moreover: (a) $\Phi'(r) \leq K/(2\gamma_{\min})$; (b) Φ' is strictly decreasing in r ; (c) $\Phi'(0^+) = K/K_c$.

Proof of (b). The integrand in (4) has the form $K\gamma/(D(\gamma + D))$ where $D = \sqrt{\gamma^2 + K^2 r^2}$. As r increases, D increases, so both D and $\gamma + D$ increase, making the integrand strictly decrease for each ω . Hence $\Phi'(r)$ is strictly decreasing. $\square$

Theorem 2.4 (Fixed point uniqueness). *For $K > K_c$, the equation $\Phi(r) = r$ has a unique solution $r^* \in (0, 1)$. Moreover, $\Phi(r) > r$ for $r < r^*$ and $\Phi(r) < r$ for $r > r^*$.*

3 Critical Algebraic Decay

3.1 Cubic drop estimate

The key ingredient is a lower bound on $r - \Phi(r)$ at criticality.

Theorem 3.1 (Cubic drop at K_c). *Let $K = K_c$ and $0 < r \leq 1$. Then*

$$r - \Phi(r) \geq \frac{K^3 r^3}{2\gamma_{\max}(2\gamma_{\max} + K)^2}. \quad (5)$$

Proof. At $K = K_c$, the identity $\int K/(2\gamma(\omega)) d\mu = 1$ holds by definition of K_c . Therefore $r = \int_{\Omega} Kr/(2\gamma(\omega)) d\mu(\omega)$. It suffices to show for μ -a.e. ω that

$$\frac{Kr}{2\gamma} - \alpha^*(\gamma, K, r) \geq \frac{K^3 r^3}{2\gamma_{\max}(2\gamma_{\max} + K)^2}.$$

Using the rationalized formula, the left side equals

$$\frac{Kr}{2\gamma} - \frac{Kr}{\gamma + D} = Kr \cdot \frac{D - \gamma}{2\gamma(\gamma + D)}$$

where $D = \sqrt{\gamma^2 + K^2 r^2}$. Since $D^2 - \gamma^2 = K^2 r^2$, we have $D - \gamma = K^2 r^2 / (D + \gamma)$, giving

$$Kr \cdot \frac{K^2 r^2}{2\gamma(\gamma + D)^2} = \frac{K^3 r^3}{2\gamma(\gamma + D)^2}.$$

For $r \leq 1$: $D = \sqrt{\gamma^2 + K^2 r^2} \leq \sqrt{(\gamma + K)^2} = \gamma + K$, so $\gamma + D \leq 2\gamma + K \leq 2\gamma_{\max} + K$ and $\gamma \leq \gamma_{\max}$. Hence

$$\frac{K^3 r^3}{2\gamma(\gamma + D)^2} \geq \frac{K^3 r^3}{2\gamma_{\max}(2\gamma_{\max} + K)^2}.$$

Integrating against μ (using $\int K/(2\gamma) d\mu = 1$) completes the proof. $\square$

3.2 Algebraic decay via reciprocal growth

Theorem 3.2 (Critical algebraic decay). *Let $K = K_c$, $0 < r_0 < 1$, and suppose $K^3 r_0^2 < 2\gamma_{\max}(2\gamma_{\max} + K)^2$. Define $B = K^3 r_0^2 / (\gamma_{\max}(2\gamma_{\max} + K)^2)$ and $r_n = \Phi^n(r_0)$. Then for all $n \geq 0$,*

$$r_n \leq \frac{r_0}{\sqrt{1 + Bn}}. \quad (6)$$

In particular, $r_n = O(1/\sqrt{n})$ as $n \rightarrow \infty$.

Proof. Set $c = K^3 / (2\gamma_{\max}(2\gamma_{\max} + K)^2)$ so that $r - \Phi(r) \geq cr^3$ by Theorem 3.1, and $B = 2cr_0^2$.

We first verify $\Phi(r) < r$ for all $r \in (0, 1]$ when $K = K_c$. Since $\Phi(0) = 0$ (as $\alpha^*(\gamma, K, 0) = 0$) and $\Phi'(0^+) = K/K_c = 1$ by Theorem 2.3(c), while $\Phi'(r) < 1$ for all $r > 0$ by Theorem 2.3(b), we have $\Phi(r) = \int_0^r \Phi'(s) ds < r$ for all $r > 0$. Thus the iterates satisfy $0 < r_{n+1} < r_n \leq r_0$, ensuring $cr_n^2 \leq cr_0^2 < 1$ throughout.

We prove the equivalent statement

$$r_n^2(1 + Bn) \leq r_0^2 \quad (7)$$

by induction on n . The base case $n = 0$ is immediate.

For the inductive step, assume (7) holds for n . From the cubic drop, $r_{n+1} \leq r_n(1 - cr_n^2)$. Squaring:

$$r_{n+1}^2 \leq r_n^2(1 - cr_n^2)^2.$$

The algebraic inequality $(1 + 2t)(1 - t)^2 \leq 1$ for $t \in [0, 1]$, verified by $1 - (1 + 2t)(1 - t)^2 = t^2(3 - 2t) \geq 0$, applied with $t = cr_n^2 \in [0, 1]$ gives

$$r_{n+1}^2(1 + 2cr_n^2) \leq r_n^2. \quad (8)$$

Now the division chain. From the inductive hypothesis, $1 + Bn \leq r_0^2/r_n^2$. We compute:

$$\frac{r_0^2}{r_n^2}(1 + 2cr_n^2) = \frac{r_0^2}{r_n^2} + \frac{r_0^2}{r_n^2} \cdot 2cr_n^2 = \frac{r_0^2}{r_n^2} + 2cr_0^2 = \frac{r_0^2}{r_n^2} + B.$$

Therefore:

$$1 + B(n + 1) = (1 + Bn) + B \leq \frac{r_0^2}{r_n^2} + B = \frac{r_0^2}{r_n^2}(1 + 2cr_n^2) \leq \frac{r_0^2}{r_{n+1}^2},$$

where the last inequality uses (8). This establishes (7) for $n + 1$.

From (7): $r_n^2 \leq r_0^2/(1 + Bn)$, hence $r_n \leq r_0/\sqrt{1 + Bn}$ since $r_n > 0$. $\square$

Remark 3.3. The $O(1/\sqrt{n})$ rate is the generic rate for iteration of a map with cubic tangency to the identity at a fixed point (cf. the continuous analogue $dr/dt = -cr^3$ giving $r(t) \sim 1/\sqrt{t}$). The contribution here is the explicit constant B depending on the frequency distribution via $\gamma_{\max}$.

4 Geometric Convergence Above K_c

Theorem 4.1 (Geometric convergence). *Let $K > K_c$ with fixed point r^* . Let $r_0 \in (0, r^*)$ and define $L_0 = \Phi'(r_0)$ as in (4). If $L_0 < 1$, then for all $n \geq 0$:*

$$r^* - r_n \leq L_0^n (r^* - r_0). \quad (9)$$

Proof. By Theorem 2.3(b), Φ' is strictly decreasing in r . Since $r_n \geq r_0$ for all n (the iterates increase toward r^* by Theorem 2.4), we have $\Phi'(r_n) \leq \Phi'(r_0) = L_0 < 1$. By the mean value theorem,

$$r^* - r_{n+1} = \Phi(r^*) - \Phi(r_n) \leq \Phi'(r_0) (r^* - r_n) = L_0 (r^* - r_n) \leq L_0^{n+1} (r^* - r_0). \quad \square$$

Corollary 4.2 (Critical slowing down). *Since Φ' is decreasing (Theorem 2.3(b)) and $\Phi'(0^+) = K/K_c$, the condition $\Phi'(r_0) < 1$ holds for all r_0 above a threshold that vanishes as $K \rightarrow \infty$. As $K \downarrow K_c$, the best available rate $\Phi'(r^*) \rightarrow 1$, recovering the algebraic regime of Section 3.*

5 Stability on the Complex Ott–Antonsen Manifold

The Ott–Antonsen reduction [8] yields an infinite-dimensional ODE on the open unit disk parametrized by $\omega \in \Omega$. We present two complementary stability results: an unconditional pair bound for the real scalar reduction, and an axiom-based convergence theorem for the general complex case. In this section we specialize to $\Omega \subseteq \mathbb{R}$ (so that ω denotes a real frequency) with density $g(\omega) \geq 0$, $\int g d\mu = 1$. The setup of Section 2 corresponds to $g \equiv 1$ with $\gamma(\omega) = |\omega|$.

5.1 The complex OA equation

The Ott–Antonsen equation on the unit disk is

$$\dot{z}_\omega = -i\omega z_\omega + \frac{K}{2}(\bar{\eta} - \eta z_\omega^2), \quad \eta = \int_{\Omega} \bar{z}_\omega g(\omega) d\mu(\omega), \quad (10)$$

where $g : \Omega \rightarrow \mathbb{R}_{\geq 0}$ with $\int g d\mu = 1$. The open unit disk $|z_\omega| < 1$ is forward-invariant: $\frac{d}{dt}|z_\omega|^2 = K \operatorname{Re}(\eta z_\omega)(1 - |z_\omega|^2)$. The equilibrium z_ω^* satisfies $|z_\omega^*| = \alpha^*(|\omega|, K, r^*)$ with $r^* = \eta^* > 0$ (real on the symmetric subspace).

5.2 Dietert energy identity

Theorem 5.1 (Ψ -monotonicity). *Define the Dietert energy*

$$\Psi(t) = - \int_{\Omega} \log(1 - |z_\omega(t)|^2) g(\omega) d\mu.$$

Then along solutions of (10),

$$\frac{d\Psi}{dt} = K|\eta(t)|^2 \geq 0. \quad (11)$$

In particular, Ψ is non-decreasing: once $\Psi(0) > 0$ (i.e., the oscillators are not all at $z_\omega = 0$), the system cannot return to the fully incoherent state $z_\omega \equiv 0$.

The proof (`ComplexOAEnergy.lean`, 0 sorry) uses disk invariance:

$$\frac{d}{dt}(-\log(1 - |z_\omega|^2)) = K \operatorname{Re}(\eta z_\omega).$$

Integrating over Ω and using $\int z_\omega g d\mu = \bar{\eta}$ gives $d\Psi/dt = K \operatorname{Re}(\eta \bar{\eta}) = K|\eta|^2$.

5.3 Pair bound (real scalar case)

Lemma 5.2 (Pair bound). *On the symmetric subspace ($z_\omega \in \mathbb{R}$, $g(\omega) = g(-\omega)$), let $V(t) = \int_{\Omega} (z_\omega(t) - z_\omega^*)^2 g(\omega) d\mu$. Then $dV/dt \leq 0$.*

The proof (`L2Lyapunov.lean`, 0 sorry) applies the Fubini identity to express dV/dt as a double integral: for each pair (ω_1, ω_2) , the integrand equals $-K \cdot Q(\omega_1, \omega_2)$ where $Q \geq 0$ is a sum-of-squares expression whose non-negativity uses $0 < z_\omega^* < 1$. The pointwise SOS decomposition is the algebraic core (~ 250 lines).

Remark 5.3. The pointwise pair bound is specific to real z . For complex z , a counterexample shows the pointwise bound fails (`ComplexPairBoundProof.lean`); the integral version remains an open question.

5.4 Stability theorem

Theorem 5.4 (OA stability). *Let $V(t) = \int |z_\omega(t) - z_\omega^*|^2 g d\mu$ and $r(t) = \int z_\omega(t) g d\mu = \text{Re } \eta(t)$ on the symmetric subspace.*

- (a) *(Real scalar, 0 axioms) On the symmetric subspace with $V(0) < (r^*)^2$: $V(t) \rightarrow 0$ and $r(t) \rightarrow r^*$.*
- (b) *(General, axiom-based) Under the hypotheses of the Dietert–Fernandez theorem [4]: $V(t) \rightarrow 0$ and $|\eta(t)| \rightarrow r^*$.*

Proof. For (a): V is non-increasing by Lemma 5.2 and non-negative, so $V(t) \rightarrow V_\infty \geq 0$. The basin condition $V(0) < (r^*)^2$ gives by Cauchy–Schwarz:

$$|r(t) - r^*| = \left| \int_\Omega (z_\omega(t) - z_\omega^*) g d\mu \right| \leq \sqrt{V(t)} < r^*,$$

so $r(t) \geq r^* - \sqrt{V(0)} =: \delta > 0$ for all t . With $r(t)$ bounded below, the contraction mechanism in dV/dt (from the pair bound) can be quantified: the proof decomposes V into “body” and “tail” contributions, bounds the body using $r(t) \geq \delta$ and the structure of the pair integrand, and applies a Barbalat-type argument to conclude $V(t) \rightarrow 0$. The full argument (~ 700 lines) is in `ContinuumSolvedFinal.lean`, 0 sorry.

For (b): the Dietert–Fernandez Landau damping theorem gives $V(t) \rightarrow 0$ directly. By Cauchy–Schwarz applied to $\eta(t) - \eta^*$:

$$|\eta(t) - r^*|^2 = \left| \int_\Omega (\bar{z}_\omega(t) - \bar{z}_\omega^*) g d\mu \right|^2 \leq V(t) \rightarrow 0,$$

hence $|\eta(t)| \rightarrow r^*$. □

6 From OA to the Full Kuramoto PDE

Remark 6.1 (Connection to the full Kuramoto PDE). The OA stability results (Sections 2–5) concern dynamics on the OA manifold (an ODE system parametrized by ω). For the full kinetic Kuramoto equation, Dietert and Fernandez [4] prove that under regularity conditions on g (Sobolev decay of $\hat{g}$) and linear stability of the partially locked state, solutions starting δ -close in weighted Sobolev norm satisfy $V(t) \rightarrow 0$ with algebraic decay. In the formalization, this uses two axioms: `dietert_landau_damping_2017` (Landau damping gives $V(t) \rightarrow 0$ on the OA manifold) and `oa_manifold_attractivity` (full PDE solutions converge to the OA manifold). The Cauchy–Schwarz step $|\eta(t)| \rightarrow r^*$ is machine-checked. This cleanly separates the verified OA theory from the analytic arguments (whose formalization requires Sobolev space theory not yet available in Lean/Mathlib).

7 Formal Verification

The proof is implemented in 87 Lean 4 files ($\sim 27,000$ lines) built against Mathlib v4.30.0-rc1, containing 0 uses of `sorry` and 2 explicit axioms (both encoding the Dietert–Fernandez

theorem: one for Landau damping on the OA manifold, one for OA attractivity from the full PDE). The source is available at <https://github.com/taejun-song/kuramoto-lean>.

The dependency graph has three branches: (1) **Real scalar**: Defs $\rightarrow$ EquilibriumFormula $\rightarrow$ SelfConsistencyContraction $\rightarrow$ IterationConvergence $\rightarrow$ QuantitativeRefinements; (2) **Complex OA**: ComplexOA $\rightarrow$ ComplexOAEnergy $\rightarrow$ ComplexOAPairBound $\rightarrow$ ComplexOAStability $\rightarrow$ ComplexOAConvergence; (3) **PDE wiring**: FullKuramotoTheorem $\rightarrow$ KuramotoEndToEnd.

Key formalization challenges include: nonlinear arithmetic requiring `nlinarith` with `sq_nonneg` witnesses for polynomial inequalities; division chains structured as explicit `calc` blocks with `mul_div_cancel`⁰ to satisfy Lean’s kernel; and measure-theoretic integration via `integral_mono` with integrability from the uniform bound $|\alpha^*| < 1$.

8 Discussion

The algebraic decay rate $r_n = O(1/\sqrt{n})$ at $K = K_c$ makes explicit the critical slowing down that is qualitatively well-understood: at criticality, Φ has third-order tangency with the identity ($\Phi'(0^+) = 1$, $\Phi''(0^+) = 0$), and the cubic nonlinearity governs the decay. The constant $B = K^3 r_0^2 / (\gamma_{\max}(2\gamma_{\max} + K)^2)$ captures the distribution-dependence of this rate.

The basin condition $V(0) < (r^*)^2$ in Theorem 5.4(a) is required for the Barbalat argument but not fundamental: the Dietert–Fernandez axioms (Theorem 5.4(b)) gives $V \rightarrow 0$ from Sobolev-small initial data without passing through the basin. The Ψ -monotonicity prevents return to incoherence but does not by itself force V into the basin. Extending the pair bound from the real scalar to the full complex case remains an open problem (the pointwise bound fails; the integral version is conjectured positive).

Open problems

1. **Eliminate the axioms**: formalize Dietert–Fernandez Landau damping in Lean 4 (requires Sobolev spaces and resolvent estimates).
2. **Sharp constants**: is the constant B optimal, or can it be improved to depend on $\gamma_{\min}$?
3. **Subcritical decay**: prove exponential decay $r_n \rightarrow 0$ for $K < K_c$ with explicit rate (partial results exist in the formalization).

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Data availability

The Lean 4 source code is available at <https://github.com/taejun-song/kuramoto-lean>.

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